

DUAL-CORE OPTICAL MOTION TESTBED

FIRMWARE, HOST SOFTWARE & RESULTS

STM32H747 dual-core control architecture, 1 kHz encoder loop, telemetry integrity, automated characterization, and measured results

ENGINEERING REQUIREMENTS DOCUMENT | REVISION 7.1 | 13 AUGUST 2026

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Document scope	Dual-core firmware architecture, host telemetry, safety behavior, verification, and measured motion performance
Evidence basis	Private firmware review, GUI recordings, measured plots, test evidence, and controlled engineering analysis
Distribution	Public portfolio release; implementation-level IP and reproduction instructions withheld
DOCUMENT PURPOSE Defines public functional, performance, interface, safety, verification, and acceptance requirements for the dual-core firmware and host test system while preserving implementation-level intellectual property.	

Functional Requirements and Engineering Context

Public release intent

This document is deliberately detailed about requirements, architectural decisions, operating behavior, verification, failure modes, and measured performance. It does not publish source code, memory addresses, linker placement, MPU/cache settings, synchronization primitives, board-specific mappings, build instructions, or any information intended to enable reproduction of the implementation.

Purpose, Scope, and System Boundary

This FRD governs the public-facing behavior of the STM32H747 dual-core motion-control subsystem and its host-side telemetry/test environment. The controlled scope includes workload separation, closed-loop encoder control, operator command/state behavior, telemetry integrity, fault handling, automated characterization, and measured performance evidence. Mechanical construction and manufacturing details remain in the companion platform FRD.

The reviewed private v0.3 firmware archive is used only as an evidence basis for public claims. Implementation-level details remain private. The current archive supports an encoder-based closed position/speed path and a separate current-sensed FOC integration path; only the behavior actually supported by the reviewed implementation is claimed here.

Boundary element	Included publicly	Withheld
Target control	Core roles, control cadence, state behavior, fault classes, command/telemetry behavior	Source, ISR/timer implementation, peripheral mappings, tuning constants, board hooks
Inter-core communication	Logical separation of command/status and continuous telemetry; integrity observability	Memory map, cache/MPU policy, barriers, synchronization code, exact buffer layout
Host/test	Live telemetry, logging purpose, automated characterization, retained evidence	Source code, deployment/configuration procedures, internal file formats
Verification	Measured plots, requirements traceability, known limitations, failure-mode coverage	Private benches, reproduction procedures, internal debugging infrastructure

1. System Role and Architectural Decision

The project uses the STM32H747 as a heterogeneous dual-core controller, with a 400 MHz Cortex-M7 and a 200 MHz Cortex-M4. The architectural decision was not driven by the modest public telemetry rate. It was driven by the need to preserve deterministic motion-generation headroom while communication, telemetry, diagnostics, and test orchestration continue concurrently.

In the project history, the highest timing pressure came from operating the drivetrain at fine microstep resolution while commanding higher mechanical speed. Fine microstepping increases the required STEP-event frequency for a given output speed; moving variable-latency communication and telemetry work away from the timing-critical control path protects pulse regularity and leaves more execution margin for motion control. The current private v0.3 branch evolves that same separation toward encoder-controlled three-phase actuation and a fixed-rate control task.

Architecture trade-off

A single 400 MHz M7 can easily process 100 Hz telemetry. Dual-core operation is justified here by workload isolation and timing margin, not by the telemetry bandwidth itself. This distinction is retained explicitly so the portfolio does not overstate the need for multicore processing.

Decision factor	Single-core consequence	Dual-core benefit
Timing-critical motion work	Shares scheduling and interrupt budget with communications and diagnostics	Control workload can be isolated from variable-latency service work
Higher motion-event rates	More sensitivity to blocking host/telemetry activity	Additional execution margin and clearer ownership boundaries
Telemetry/test expansion	Feature growth increases coupling to control timing	Growth can be absorbed on the service side without redesigning the control path
Complexity	Simpler implementation and debugging	Requires explicit IPC, state ownership, coherency, startup, and fault contracts

2. Functional Requirements

ID	Requirement	Engineering rationale	Verification
FW-FR-001	The subsystem shall separate timing-critical control work from communication/telemetry service work.	Prevents nondeterministic host/service activity from consuming the control execution budget.	Architecture review and concurrent-load test.
FW-FR-002	The control path shall execute at a fixed nominal cadence of 1 kHz in the reviewed integration.	Deterministic state estimation and actuation require bounded sample timing.	Target timing observation and log review.
FW-FR-003	The subsystem shall expose commanded state, measured state, operating state, and fault state to the host.	Operators need direct observability of whether actual behavior follows requested behavior.	GUI/telemetry demonstration.
FW-FR-004	Inter-core data transfer shall make loss, stalling, overflow, and sequence discontinuity observable.	Silent corruption or silent data loss is unacceptable in characterization data.	Counter/sequence fault-injection test.
FW-FR-005	PWM/actuation outputs shall be inhibited outside the permitted running state or whenever a fault is active.	The software safe state must dominate commanded operation.	State-transition and fault-response test.
FW-FR-006	Position commands shall be constrained by configured travel limits and invalid encoder/timing conditions shall prevent active output.	Motion control must remain bounded by validated operating conditions.	Boundary-command and invalid-feedback tests.

FW-FR-007	The host shall support repeatable characterization with retained telemetry and analysis context.	Results must be traceable to the commanded test rather than exist only as screenshots.	Automated test run and retained-data review.
FW-FR-008	Public claims shall distinguish the commissioned encoder-control path from the uncommissioned current-sensed FOC integration path.	Prevents the portfolio from claiming performance not supported by the reviewed source and evidence.	Document/source reconciliation.

3. Control, State, and Safety Behavior

The reviewed v0.3 control architecture is built around an encoder-based position/speed path. The control state model distinguishes disabled, homing, armed, running, and fault conditions. Energized actuation is permitted only when the controller is in the running state and no fault condition is active. All other states resolve to a disabled or neutral output condition.

Behavior	Public requirement	Reason
Fixed-rate control	Nominal 1 kHz execution for the reviewed encoder-control path	Provides a bounded update interval for estimation and closed-loop behavior.
Encoder validity	Fresh, plausible feedback is required for active closed-loop operation	Prevents stale or invalid position feedback from being treated as current state.
Timing supervision	Unexpected control timing is treated as a fault condition	Avoids applying control laws with invalid sample intervals.
Travel/speed supervision	Commanded and measured motion remain within configured operating limits	Constrains the controller to the verified mechanical envelope.
Output gating	Actuation is enabled only in a valid running state with no active fault	Ensures fault handling dominates normal command execution.
Watchdog/fault recovery	Faults require deliberate recovery before motion resumes	Avoids automatic re-energization after abnormal behavior.

4. Dual-Core IPC and Shared-Memory Correctness

The dual-core subsystem uses two logical communication classes: low-rate command/status exchange and a bounded shared-memory path for continuous motion records. That separation keeps command semantics independent of streaming telemetry and allows the service core to consume data without forcing every record through the command channel.

The important engineering risk is not bandwidth; it is correctness at the ownership boundary. The Cortex-M7 cache architecture means shared memory can become stale, partially published, or observed out of order unless the implementation defines a coherent memory policy and synchronization contract. Cache coherency is therefore treated as a system-level requirement, not an implementation footnote.

Integrity concern	Required property	Public verification statement
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Cache coherency	A consumer shall not act on stale or partially published shared data.	Concurrent-core stress testing and consistency checks are required.
Publication ordering	Record contents shall become valid before ownership/index state advertises the record.	Sequence and consistency tests verify complete-record observation.
Buffer bounds	Metadata plus retained records shall remain within the reserved object boundary.	Compile-time size contracts guard against layout regression.
Sequence continuity	Loss or overwrite shall be observable rather than silent.	Sequence-gap and drop accounting are part of verification.
Timestamp rollover	Elapsed-time reconstruction shall remain valid across finite-width timestamp wrap.	Host/analysis rollover tests are part of the validation set.
Version compatibility	Protocol revisions shall remain representable and explicitly checked.	Interface-version validation is required before accepting messages.

Corrected boundary defect

Engineering review identified an earlier shared-buffer sizing defect in which record capacity consumed the complete reservation without leaving room for metadata. The private implementation was corrected and compile-time fit checks were added. Exact object layout and memory placement are intentionally omitted from this public release.

5. Host Application, Telemetry, and Test Workflow

The host application provides the operator-facing layer for connection state, command entry, live position and speed monitoring, scripted characterization, data retention, and analysis. The host is treated as a test/observability subsystem rather than a hard real-time controller; time-critical safety behavior remains on the target.

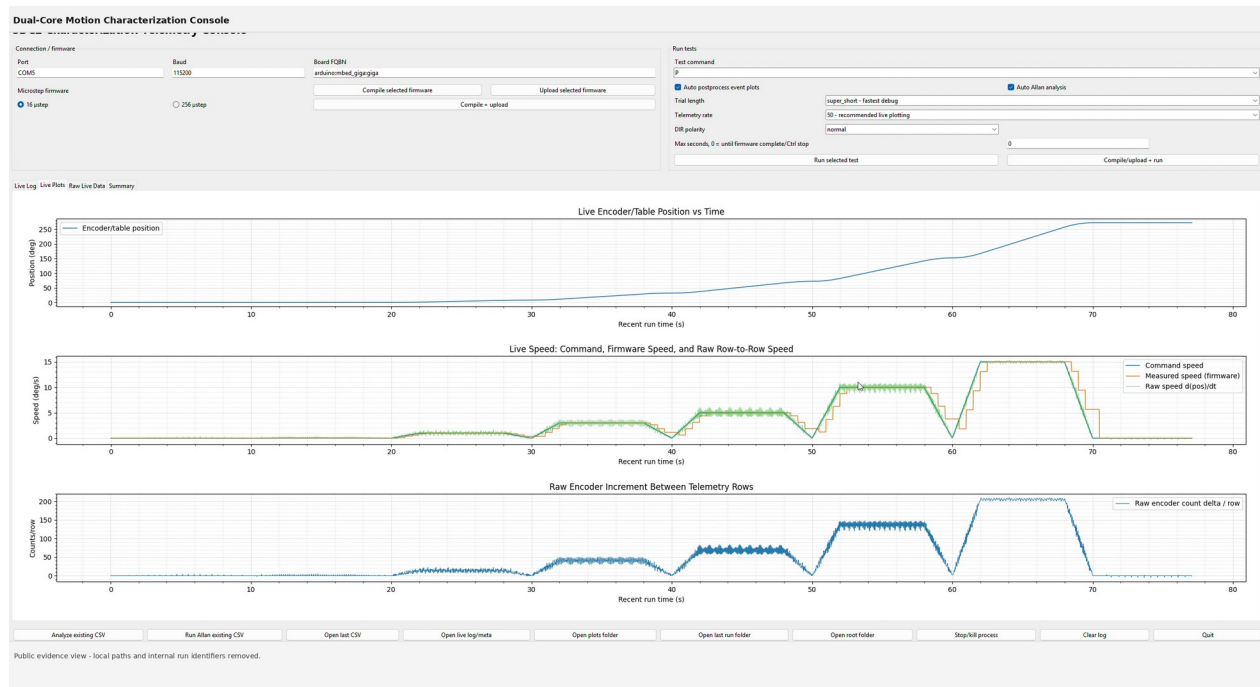


Figure 1. Host telemetry interface showing commanded/measured motion and live encoder-derived data.

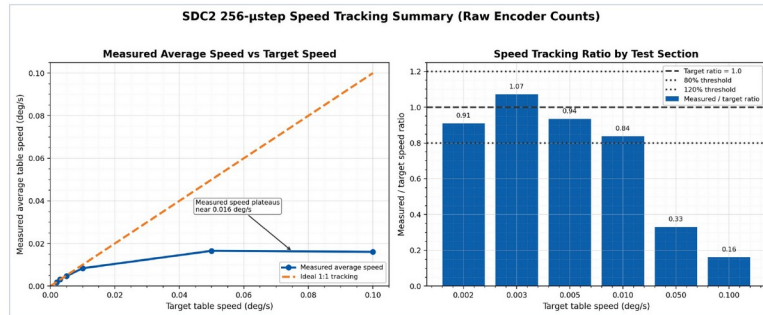
Host function	Required behavior	Evidence retained
Connection/status	Make target availability and operating state visible	GUI state and run records
Live telemetry	Display commanded and measured motion without hiding fault/integrity state	Captured GUI plots
Automated tests	Execute repeatable motion profiles and retain run identity/context	Characterization datasets and plots
Logging	Preserve analysis-ready measurements with enough context to trace each run	CSV/metadata evidence
Recovery	Differentiate a communication interruption from a target fault or invalid run	Observed reconnect/fault behavior

6. Measured Motion Characterization

The public evidence set retains measured motion behavior rather than implementation internals. The useful ultra-low-speed range, constant-speed tracking, transient response, and closed-loop sine-response tests collectively show how the integrated control/test system behaves across different operating conditions. These figures are evidence, not a substitute for the withheld implementation.

SDC2 ONLY | 256 MICROSTEP BASELINE

Speed tracking summary: usable ultra-slow range, clear high-speed ceiling



Main result

- 0.002-0.01 deg/s tracks in the correct range at the current 256 microstep setting.
- Best tracking occurs around 0.003-0.005 deg/s; 0.01 deg/s is usable but below target.
- 0.05 and 0.10 deg/s fall similarly and plateau near 0.016 deg/s actual speed.
- Interpretation: higher commanded speeds likely require lower microstepping and/or controller tuning.

Numerical baseline

- 0.002 → 0.001818 deg/s, ratio 0.909
- 0.003 → 0.003214 deg/s, ratio 1.071
- 0.005 → 0.004676 deg/s, ratio 0.935
- 0.010 → 0.008378 deg/s, ratio 0.838
- 0.050 → 0.016560 deg/s, ratio 0.331
- 0.100 → 0.016071 deg/s, ratio 0.161

SDC2 Encoder-Based Speed Tracking

Raw encoder-event analysis | no filtering or grouped averaging

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Figure 2. Low-speed tracking summary showing the usable ultra-slow range and the observed high-speed ceiling of the tested configuration.

The low-speed data show the strongest tracking in the approximate 0.003-0.005 deg/s region. At higher requested speed in the tested fine-resolution configuration, the measured response approaches a ceiling rather than continuing linearly. That behavior is one of the reasons the architecture work focused on preserving motion-generation headroom while retaining fine motion resolution.

Commanded speed	Measured speed	Interpretation
0.002 deg/s	0.001818 deg/s	Low-speed operation demonstrated; quantization and estimator effects are increasingly visible.
0.003 deg/s	0.003214 deg/s	Strong agreement in the tested ultra-slow region.
0.005 deg/s	0.004676 deg/s	Strong agreement with modest negative bias.
0.010 deg/s	0.008378 deg/s	Tracking remains usable but error increases.
0.05-0.10 deg/s	Approximately 0.016 deg/s plateau in this configuration	Demonstrates a configuration-specific ceiling rather than a universal system limit.

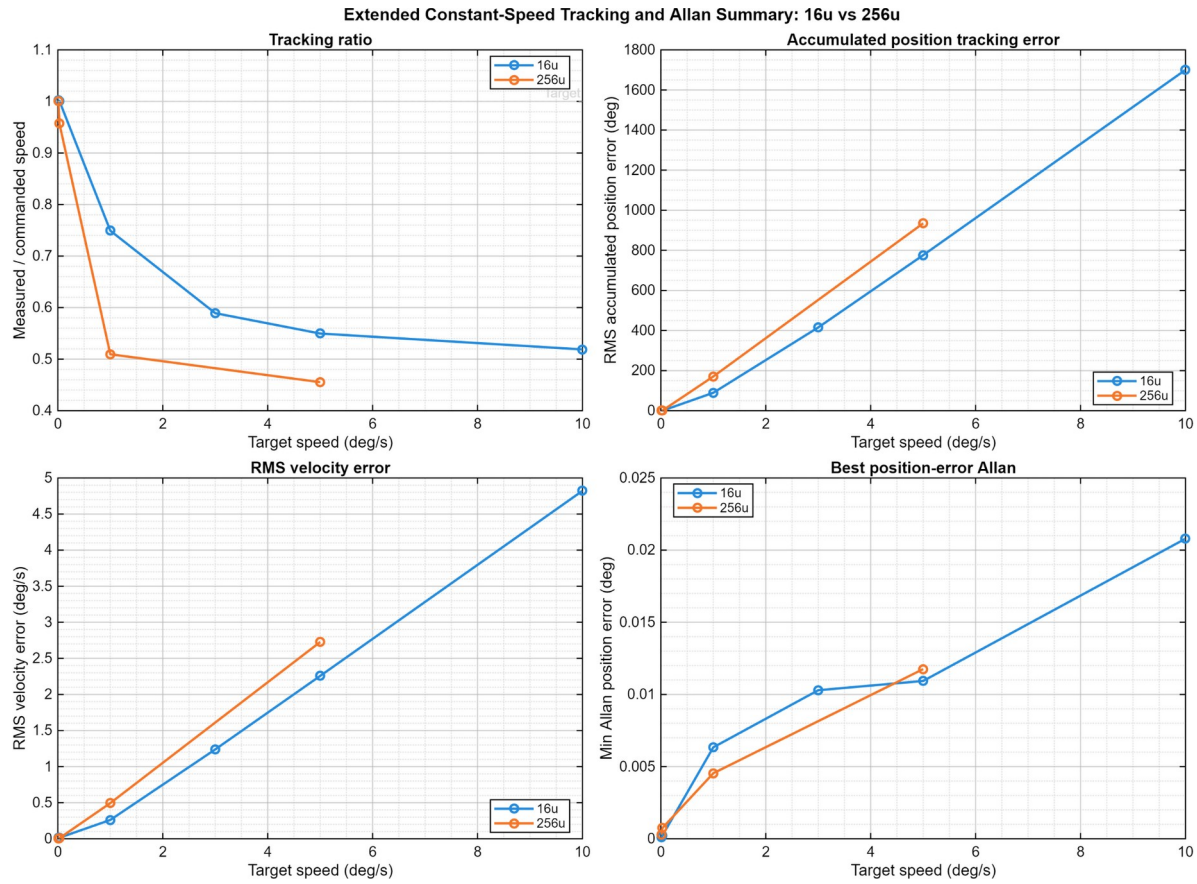


Figure 3. Extended constant-speed tracking and stability characterization.

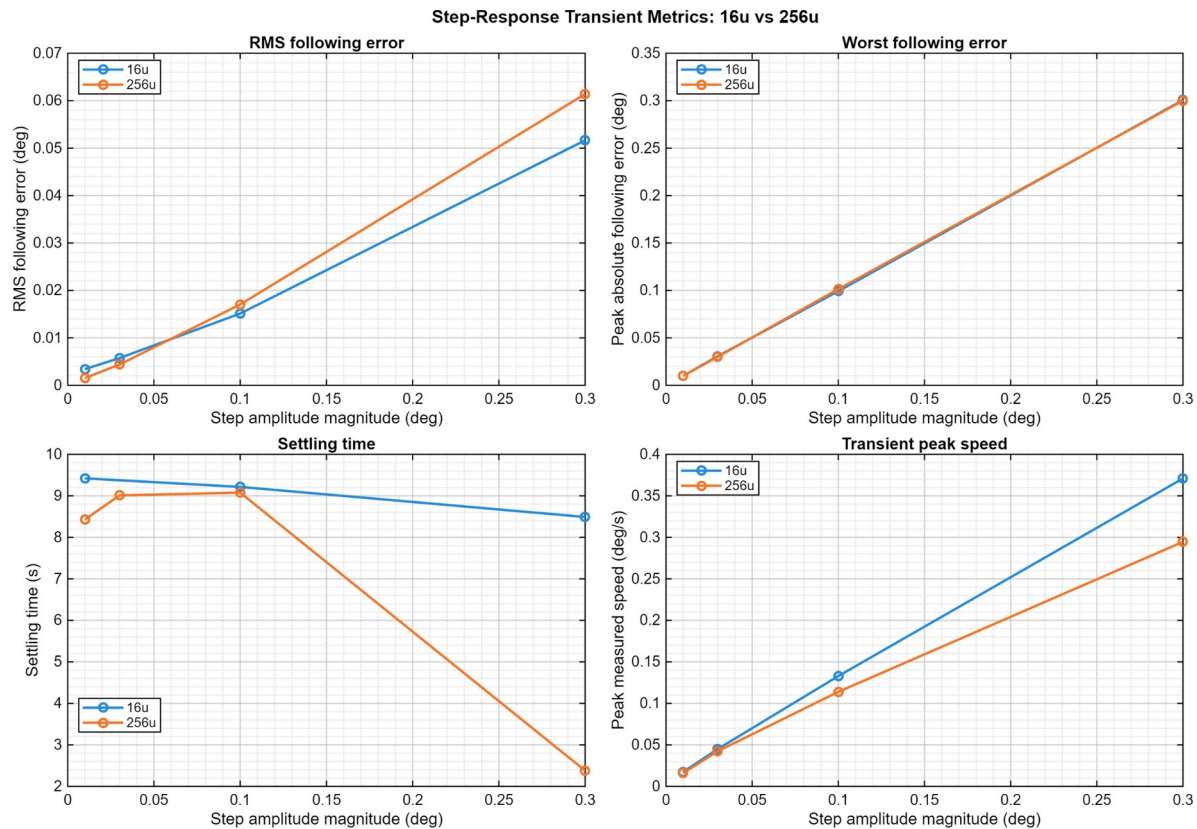


Figure 4. Step-response characterization using multiple transient error and tracking metrics.

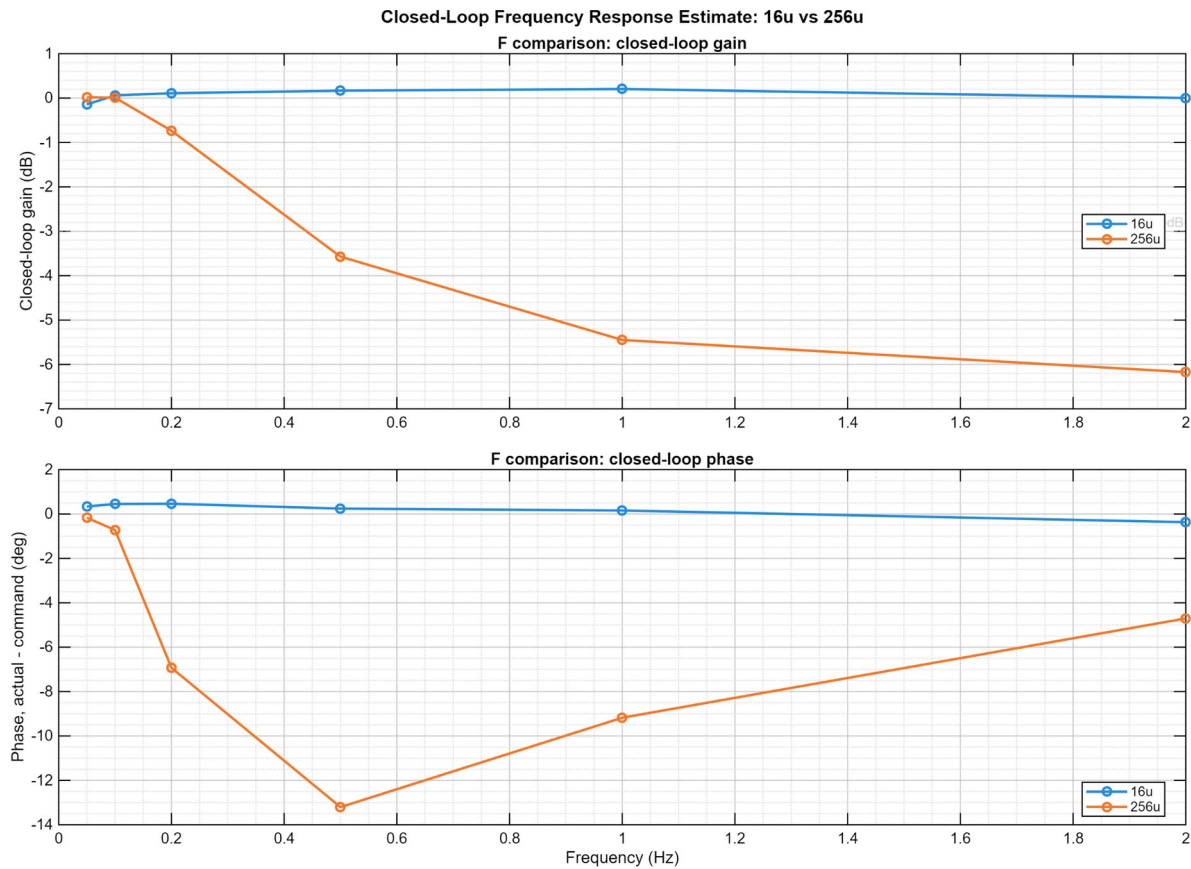


Figure 5. Closed-loop sine-response estimate across the tested 0.33-2 Hz range.

The frequency-response evidence supports reporting behavior across the tested frequencies. It does not independently establish a formal -3 dB bandwidth because the available data do not demonstrate a complete, well-resolved crossing. Public claims are intentionally limited to the measured range.

7. Current Firmware Maturity and FOC Boundary

The private v0.3 archive contains two distinct motor-control paths. The deployable path supported by the available feedback is an encoder-based closed position/speed controller that produces a sensed rotating voltage command. It is not current-regulated field-oriented control. A separate current-sensed FOC integration path is present for future commissioning once calibrated phase-current feedback and the required power-stage integration are available.

Capability	Current public disposition
Dual-core command/telemetry architecture	Implemented and reviewed at the application layer.
Encoder-based position/speed control	Implemented in the private v0.3 integration and treated as the current control path.
1 kHz control cadence	Defined for the reviewed integration; fixed-rate hardware-triggered execution is the target for energized operation.
Current-regulated d/q FOC	Integration path exists, but commissioned platform operation is not claimed.
Power-stage/peripheral-specific implementation	Private and outside the public release boundary.

8. Verification and Traceability

Requirement	Verification evidence	Disposition
FW-FR-001	Architecture review and concurrent workload analysis	Satisfied at architecture level
FW-FR-002	Reviewed control-task cadence and timing requirement	Implemented in reviewed integration
FW-FR-003	GUI captures and logged commanded/measured state	Demonstrated
FW-FR-004	Integrity counters, sequence continuity, and fault-observability requirements	Implemented/verified privately; public method stated
FW-FR-005	State/fault gating review and safe-output requirement	Implemented in reviewed integration
FW-FR-006	Travel/speed/invalid-feedback boundary behavior	Defined and privately verified
FW-FR-007	Automated characterization plots and retained data products	Demonstrated
FW-FR-008	Source-to-document reconciliation of encoder path versus FOC path	Satisfied

9. Limitations and Public Release Controls

- No source code, Simulink/model files, memory maps, linker placement, cache/MPU configuration, synchronization code, pinouts, schematics, PCB files, BOMs, or setup/flash procedures are included.
- Measured performance is specific to the tested hardware/configuration and is not presented as a universal controller limit.
- Current-sensed FOC commissioning is not claimed by the reviewed v0.3 evidence.
- The public coherency discussion defines the correctness requirement and verification intent, not the private implementation method.
- Formal EMC, environmental, functional-safety, and production qualification are outside the scope of this portfolio FRD.

Document Control

Revision	Date	Description
7.1	13 August 2026	Restored the established v5 FRD template; expanded architecture rationale, control-state behavior, shared-memory correctness requirements, firmware maturity, measured evidence, and traceability while removing reproduction-level IP.